

Quantum Computing

Lecture notes from my graduate level Quantum Computing class taught by professor Lijun Chen at the University of Colorado Boulder. This course covers a range of foundational topics in quantum computation and information including: basics to quantum mechanics, entanglement, quantum optimization and machine learning, reliable and scalable quantum computing, and quantum communication and networking. There is no official textbook we are using, but rather a bunch of lecture slides from various undergrad and grad courses.

Disclaimer: This document will contain mistakes such as typos, errors, etc. These were taken during the process of learning a new subject, so don't take these notes too seriously.

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1 Basics of Quantum Mechanics

1.1 Bit vs qubit

A bit is the basic unit of computation that takes a value of a 0 or a 1. Think of it as a two-state system: on or off, low or high etc.

The quantum bit (qubit) is a basic unit of information that takes in two-dimensional complex vector space.

$$|\psi\rangle = a|0\rangle + b|1\rangle, |a|^2 + |b|^2 = 1 \quad (1)$$

- $|0\rangle$ and $|1\rangle$: elements of an orthonormal basis
- $a, b \in \mathbb{C}$ are the probability amplitudes which must satisfy $\sum_i |a_i|^2 = 1$
- if measured onto the basis, $|0\rangle$ is obtained with a probability $|a|^2$ and $|1\rangle$ with probability $|b|^2$

1.2 Basics of Quantum Mechanics

- how do we represent a quantum state?
- how to represent an observable?
- how does a quantum system evolve over time?
- what operations can we do on a quantum system?
- how do we combine quantum systems?

1.2.1 States

A state is a complete description of a physical system. A state in quantum mechanics is defined as a unit vector in Hilbert space, denoted by $|\psi\rangle$ (Dirac's ket notation).

- a vector space over complex numbers $\mathbb{C}$
- **Inner Product:** $\langle\psi|\phi\rangle$ that maps an ordered pair of vectors to $\mathbb{C}$ and that has these properties:
 - Positivity: $\langle\psi|\psi\rangle > 0$ for $|\psi\rangle \neq 0$
 - Skew symmetry: $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$
- **Normalization:** $\| |\psi\rangle \| = \sqrt{\langle\psi|\psi\rangle}$
- **Global phase:** $|\psi\rangle$ and $e^{ia}|\psi\rangle, a \in \mathbb{R}$ are indistinguishable

Suppose there are N distinguishable states. A quantum state $|\psi\rangle$ is a *superposition* of these N states:

$$|\psi\rangle = a_1|1\rangle + a_2|2\rangle + \dots + a_n|N\rangle \quad (2)$$

We can think of $|\psi\rangle$ as the unit vector:

$$\begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_N \end{bmatrix}$$

With the conjugate transpose:

$$\langle \psi | = (a_0^*, a_1^*, \dots, a_n^*)$$

So, for the two states $|\psi\rangle = a_n|N\rangle$ and $|\phi\rangle = b_n|B_N\rangle$, their *inner product* is:

$$\langle \psi | \phi \rangle = a_1^* b_1 + \dots + a_n^* b_n \quad (3)$$

The *outer product* (projection to $|\psi\rangle$) corresponds to:

$$\begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{bmatrix} [b_1^*, b_2^*, \dots, b_n^*]$$

1.2.2 Observable

A property of a physical system that can be measured. It's defined mathematically as a self-adjoint operator on Hilbert space

$$A = A^\dagger \quad (4)$$

- can be diagonalized:

$$A = \sum_n a_n E_n \quad (5)$$

with a_n being eigenvalues of A , and E_n is the corresponding orthogonal projection onto the space of eigenvectors with eigenvalues a_n .

- self-adjoint guarantees these eigenvalues a_n are *real* because the measurements represent actual outcomes (not fake ones)
- the projections satisfy: $E_n = E_n^\dagger$ and $E_n E_m = \delta_{n,m} E_n$

What this means is:

- Hermitian (self-adjoint)
- $\delta_{n,m}$ is the **Kronecker delta** which equals 1 if $n = m$ and - if $n \neq m$.

For case $n = m$ (idempotent) applying the same projection twice does nothing.

For case $n \neq m$ $E_n E_m = 0$. Projecting onto space m and then a different space gives you nothing because the two spaces don't overlap at all. The eigenspaces are mutually perpendicular.

1.2.3 Measurement

The measurement of an observable A prepares an eigenstate $|n\rangle$ of A (state collapse):

- Probability of outcome a_n :

$$\text{Prob}(a_n) = ||E|\psi\rangle||^2 = \langle\psi|E|\psi\rangle$$

- Post-measurement state:

$$\frac{E_n|\psi\rangle}{||E_n|\psi\rangle||} \text{ (state collapse)}$$

Note: when measured again, the outcome remains the same.

- Expected value over many trials: $\sum_n a_n \text{Prob}(a_n) = \langle\psi|A|\psi\rangle$

Measuring is literally projecting the state vector onto an eigenspace and normalizing.

1.2.4 Dynamics

Dynamics describe how a state evolves over time. The time evolution of a closed system is described by a *unitary operator*

If the initial state at time t is $|\psi\rangle_t$, then the final state $|\psi\rangle_{t'}$ at time t' can be expressed as

$$|\psi(t')\rangle = U(t', t)|\psi(t)\rangle \quad (6)$$

with unitary operator satisfying

$$U^{\dagger}U = UU^{\dagger} = I \quad (7)$$

Since the unitary operator preserves inner products: $|\psi'\rangle = U|\psi\rangle$ and $|\phi'\rangle = U|\phi\rangle$, then

$$\langle\psi'|\phi'\rangle = \langle\psi|U^{\dagger}U|\phi\rangle = \langle\psi|\phi\rangle \quad (8)$$

1.2.5 Composite system

If the Hilbert space of system $\mathcal{A}$ is $\mathcal{H}$ and $\mathcal{B}$ with $\mathcal{H}_{\mathcal{B}}$, then the Hilbert space of the composite systems AB is the *tensor product* $\mathcal{H}_{\mathcal{A}} \otimes \mathcal{H}_{\mathcal{B}}$

$$\mathcal{H}_{AB} = \mathcal{H}_{\mathcal{A}} \otimes \mathcal{H}_{\mathcal{B}} \quad (9)$$

- with dimension $d_{\mathcal{A}}d_{\mathcal{B}}$ and orthonormal basis $\{|i\rangle_{\mathcal{A}} \otimes |j\rangle_{\mathcal{B}}\}$

1.2.6 Entanglement

A state like the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ cannot be factored as a tensor product $|\psi\rangle_A \otimes |\phi\rangle_B$

1.2.7 Density matrix

When you only care about subsystem A , you compute expectation values

$\langle \psi | M_A | \psi \rangle = \text{tr}(M_A \rho_A)$ which defines the reduced density matrix $\rho_A = \text{tr}_B |\psi\rangle\langle\psi|$ the *partial trace*.

Density matrices have the following properties:

- ρ is self-adjoint $\rho = \rho^\dagger$
- ρ is positive i.e. $\langle \psi | \rho | \psi \rangle \geq 0$ for any $|\psi\rangle$
- $\text{tr} \rho = 1$
- pure state if $\text{rank}(\rho) = 1$ and mixed state otherwise
- pure state if $\rho^2 = \rho$ and mixed state otherwise

1.2.8 No cloning theorem

States that no arbitrary transformation can make a copy of an arbitrary quantum state.

Take two arbitrary states $|\psi\rangle_A$ and $|\phi\rangle_A$

$$\begin{aligned}
U|\psi\rangle_A|0\rangle_E &= |\psi\rangle_A|\psi\rangle_E \\
U|\phi\rangle_A|0\rangle_E &= |\phi\rangle_A|\phi\rangle_E
\end{aligned}$$

The unitary transformation preserves the inner product

$$\langle \psi | \phi \rangle = \langle \psi | \phi \rangle^2 \tag{10}$$

But $\langle \psi | \phi \rangle = 0$ or 1 which is a contradiction unless the states were already orthogonal or identical. Thus, cloning arbitrary states is algebraically impossible.

2 Quantum Circuits